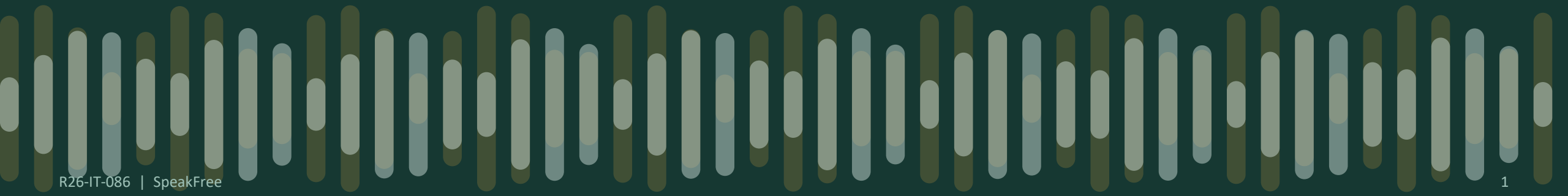
PROGRESS PRESENTATION II

SpeakFree

AI-Based Speech Disorder Screening and Therapy Assistant

Research Project R26-IT-086 - Faculty of Computing, SLIIT



Our Team

SpeakFree



P.A.S.S. Kusumsiri

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Component 1

Speech Screening & Feature Extraction



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Component 2

Alignment & Articulation Analytics



I.L.S.Tharushan

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Component 3

Multi-Disorder Severity Classification



M.D.I.Perera

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Component 4

AI Therapy Assistant & Privacy

Special Thanks,

A special thank you to Dr. Rajapaksha for their valuable medical guidance and expert insights throughout this project.

Full Name – W.R.A.R.R. Rajapaksha
SLMC Registration Number – 13510
Registration Type – MBBS (Sri Lanka)

A special thank you to my supervisor, Ms. Chathurangika Kahandawaarachchi, for her invaluable academic guidance, continuous support, and mentorship throughout this research.

A heartfelt thank you to my co-supervisor, Ms. Suranjini Silva, for her continuous encouragement and technical assistance in bringing this project to life.

Problem & Motivation

Context

Speech-language disorders (stuttering, dysarthria, and articulation impairments) affect over a billion people worldwide. In Sri Lanka, access to qualified speech-language pathologists is severely limited by clinician scarcity, geography, and the cost of sustained private therapy.

Observed Problems

- Automated tools stop at diagnosis - no therapy delivery follows
- Standard ASR performs poorly on disordered speech (60–80% error rates)
- Existing therapy apps use static, one-size-fits-all exercise plans
- Centralised voice-data platforms carry real re-identification and privacy risk

1B+

people worldwide live with a communication disability (WHO)

60–80%

word-error rate of standard ASR on disordered speech



Solution Overview

A four-component, privacy-preserving AI platform that takes a patient from first screening through personalised therapy - entirely in the browser, with no specialised hardware.

Key Features

- Noise-robust, multi-task speech capture & feature extraction
- Phoneme-level alignment & articulation analytics
- Multi-disorder, multi-severity classification
- Real-time AI therapy coaching, DAF & adaptive weekly plans

Non-Functional Goals

- Browser-based - no installation, works on any device with a mic
- Sub-300ms latency for real-time therapy coaching
- Formal privacy guarantees via differential privacy
- Modular REST/WebSocket API contracts across all 4 components

1. Screening | **2. Articulation** | **3. Severity** | **4. Therapy**

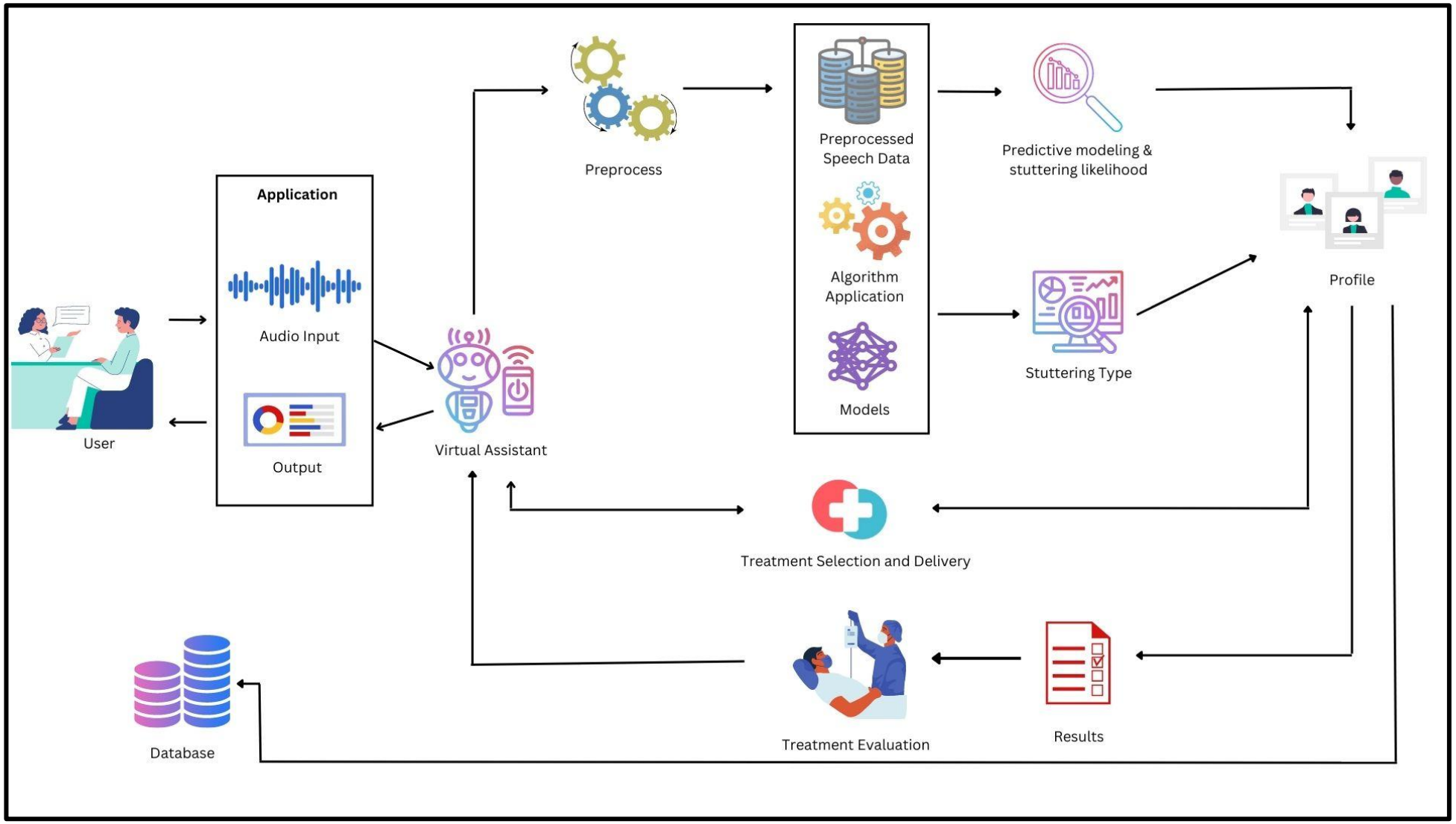
One integrated pipeline - screening feeds analytics, analytics feeds classification, classification drives therapy

Overall Research Gap

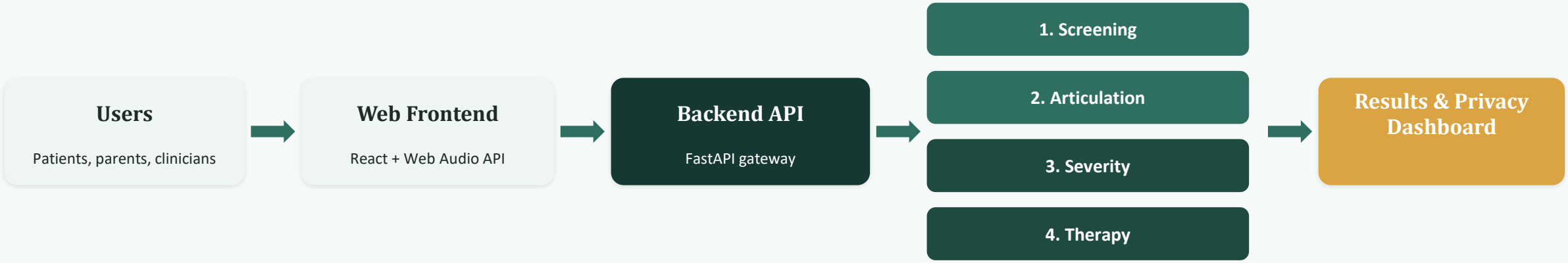
Feature	Diagnostic ASR Tools	Constant Therapy	SpeechEasy (DAF)	SpeakFree (Ours)
Noise-robust, multi-task screening	✗	✗	✗	✓
Phoneme-level articulation analytics	✓*	✗	✗	✓
Automated multi-disorder severity scoring	✓*	✗	✗	✓
Real-time adaptive therapy coaching	✗	✗	✗	✓
Session-driven adaptive weekly planning	✗	✗	✗	✓
Differential privacy / federated learning	✗	✗	✗	✓
Treatment-response biomarker tracking	✗	✗	✗	✓
One integrated end-to-end platform	✗	✗	✗	✓

** Diagnostic/ASR tools may partially cover this in isolated research settings, but not as an integrated product feature*

System Architecture & Integration



System Architecture & Integration



Integration Points

- REST API (JSON) for session, feature, and classification exchange
- WebSocket streaming for sub-300ms real-time therapy feedback
- Shared session ID links all 4 components' outputs per patient
- Audio blob upload (Component 1) - never re-uploaded downstream

Data Flow

- Component 1 → clean acoustic features, ASR confidence scores
- Component 2 → phoneme-level articulation sub-scores, boundary confidence
- Component 3 → disorder type, severity level (1–4), uncertainty estimate
- Component 4 → real-time feedback, weekly plan, privacy-budget status

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Noise-Robust Speech Screening & Feature Extraction

Capture multi-task speech in noisy, real-world browser conditions and extract clean, confidence-scored features for every downstream component.

Key Capabilities

- Browser-based multi-task capture: picture description, video narration, guided conversation
- Noise reduction + VAD silence trimming that preserves clinically meaningful pauses
- Whisper confidence-scored ASR flags low-confidence segments as dysfluency candidates
- HuBERT embeddings + Librosa acoustic features (WPM, articulation rate, energy, tempo)

Technology

React | Web Audio API | FastAPI | Whisper | HuBERT | Librosa

Output Validation

We conducted an informal review of the sample outputs with Dr. Rajapaksha, who noted no clinical inconsistencies. Please note that this serves as an initial check, not a formal validation; we still need to conduct a full study to measure statistical agreement between the system and an independent clinician.

Key Parameters -:

- *ASR Confidence*
- *Confidence*
- *Words Per Minute*
- *Energy Mean*
- *SNR (Signal-to-Noise Ratio)*
- *Articulation Rate*
- *Zero Crossing Rate (ZCR)*
- *Energy Mean*

These extraction parameters were reviewed and clinically verified by our medical supervisor, Dr. Rajapaksha.

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Alignment & Articulation Analytics

Align recognised speech against expected phoneme sequences and quantify articulation accuracy at the phoneme level.

Key Capabilities

- Forced alignment between ASR transcript and audio to locate phoneme/word boundaries
- Phoneme-level articulation accuracy scoring against target pronunciation
- Boundary-confidence estimation to flag unreliable alignments before scoring
- Structured articulation sub-scores passed to severity classification & therapy layers

Technology

Forced Alignment | wav2vec2 | Python | FastAPI

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Multi-Disorder Severity Classification

Classify disorder type and quantify severity from Component 1's acoustic features and Component 2's articulation sub-scores.

Key Capabilities

- Multi-class disorder classification — stuttering, dysarthria, articulation disorder
- Severity scoring (Level 1–4) with an explicit uncertainty estimate
- Consumes upstream feature vectors from Components 1 and 2
- Drives exercise selection downstream and mandatory SLP referral at Level 3–4

Technology

Deep Learning | PyTorch | Python | FastAPI

4

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AI Therapy Assistant & Privacy-Preserving Learning

Deliver personalised, real-time therapy while protecting sensitive voice data with formal, auditable privacy guarantees.

Key Capabilities

- Real-time pacing coach + Delayed Auditory Feedback with adaptive delay calibration
- Adaptive weekly planning via a contextual bandit (LinUCB) on session outcomes
- Differential privacy: Laplace-mechanism statistics + per-patient privacy budget ledger
- Federated learning architecture — on-device improvement, raw audio never leaves device

Technology

React | WebSockets | FastAPI | NumPy | Flower / DP-SGD design

Commercialization

Target Users / Market	Speech-language clinics, schools, and individuals in underserved regions of Sri Lanka.
Value Proposition	One browser-based platform spanning screening through therapy — no specialised hardware, no clinic visit required.
Cost & Feasibility	Built entirely on open-source AI (Whisper, HuBERT, PyTorch); runs on any smartphone or laptop with a microphone.
Adoption Plan	Pilot with SLIIT-affiliated clinics and schools → clinician-supervised trial → wider public rollout.

Estimated Costs

- Cloud hosting (containerised FastAPI + inference): ~$20–$40 / month
- Domain registration & TLS certificate: ~$15 / year
- No specialised hardware — DAF and pacing coaching run in-browser

Thank You

Questions & Discussion

